

General Relativity HW-11

PROBLEM 1. Consider Friedman equations with some normal matter ($\rho > 0$). Show that a shrinking Hubble radius implies an accelerated expansion and consequently a negative pressure. In other words:

$$\frac{d}{dt} \left(\frac{1}{aH} \right) < 0 \Rightarrow \ddot{a} > 0 \Rightarrow \rho + 3p < 0$$

SOLUTION:

I might be wrong with something, but why this problem looks like pretty trivial?

The first statement $\frac{d}{dt} \left(\frac{1}{aH} \right) < 0 \Rightarrow \ddot{a} > 0$ is obvious, just plug in $H = \frac{\dot{a}}{a} \Rightarrow \frac{d}{dt} \left(\frac{1}{aH} \right) = -\frac{\ddot{a}}{a^2} < 0 \Rightarrow \ddot{a} > 0$

By Friedmann equation,

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(p + 3\rho)$$

then

$$\ddot{a} > 0 \Rightarrow p + 3\rho < 0$$

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